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Intraoperative artificial intelligence for brain tumor surgery: from tissue diagnosis to real-time surgical guidance

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The 2021 WHO CNS tumor classification marked a fundamental shift in neuro-oncology: molecular markers rather than histological appearance now define glioma diagnosis and guide treatment. Yet, in clinical practice the molecular information that matters most to a surgeon—IDH status, 1p/19q codeletion, and methylation class—typically arrives days to weeks after the operation it was meant to inform. Artificial intelligence (AI) is beginning to close that gap. Over the past several years, a growing body of work has demonstrated that AI models can extract diagnostically relevant molecular and histological information from data sources available in the operating room, in real time, at the point of surgical decision-making. In this review, the authors provide a comprehensive overview of state-of-the-art intraoperative AI models available to neurosurgeons during brain tumor resections. They examine AI-augmented approaches to established intraoperative techniques, such as cryosection and fluorescence-guided surgery, before turning to novel technologies including nanopore sequencing and stimulated Raman histology (SRH). The authors conclude by discussing barriers to widespread clinical adoption and what lies ahead for this field. For cryosection, models such as LGNet and the Cryosection Histopathology Assessment and Review Machine (CHARM) have demonstrated strong performance on intraoperative discrimination of primary CNS lymphoma and molecular subtyping of glioma, respectively, with CHARM achieving areas under the receiver operating characteristic curve of 0.88–0.93 across glioma subtypes. In fluorescence-guided surgery, hyperspectral imaging combined with machine learning has enabled classification of tumor type, grade, and IDH mutation based on the 5-aminolevulinic acid (5-ALA) signal with up to 93% accuracy, extracting molecular information from a modality already present in the operating room. For nanopore-based methylation classification, ensemble models such as Rapid Nanopore Brain Intraoperative Classification (ROBIN) have achieved 90% concordance with the final diagnosis prospectively in under 2 hours. In SRH, foundation models for infiltration detection now outperform FLAIR, contrast enhancement, and 5-ALA fluorescence in identifying tumor margins, while molecular classification models have achieved over 90% glioma subtype accuracy in multi-institutional cohorts. At the single-cell level, morphometric profiling of SRH images has revealed nuclear features distinguishing IDH-mutant from IDH-wildtype tumors, opening new avenues for interpretability. Despite this progress, widespread adoption remains limited by the absence of randomized clinical trials demonstrating patient-level benefit. Addressing this gap and realizing the potential of multimodal AI systems that integrate imaging, pathology, and molecular data represent critical next steps for the field.

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The rise of machine learning and artificial intelligence (AI) in the neurosurgical literature since the 2010s, mirroring the pattern in the medical literature at large, has been remarkable. In recent years, AI tools and models have been shown to perform a wide array of preoperative and postoperative tasks, including information extraction from pathology reports, tumor segmentation, outcome prediction and prognostication, and postoperative complication forecasting.[1,2] Building on these advances outside the operating room, intraoperative appli-

ABBREVIATIONS 5-ALA = 5-aminolevulinic acid; AI = artificial intelligence; AUROC = area under the receiver operating characteristic curve; CHARM = Cryosection Histopathology Assessment and Review Machine; CNN = convolutional neural network; FDA = Food & Drug Administration; FL-CNN = convolutional neural network combined with second near-infrared fluorescence imaging; HGG = high-grade glioma; ICG = indocyanine green; LGG = low-grade glioma; MC = methylation class; MCF = methylation class family; PCNSL = primary CNS lymphoma; ROBIN = Rapid Nanopore Brain Intraoperative Classification; SRH = stimulated Raman histology.

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cations of AI have also emerged as a rapidly evolving area of research. One of the most exciting prospects of this new age of medicine for neurosurgeons is the integration of the vast amounts of data collected through the course of clinical care into models that directly affect treatment plans in near real time. Given a diminishing neuropathology workforce and the time pressures facing these specialists during intraoperative consultation, which has led to reports of ambiguous findings in about 10%–20% of cases, many intraoperative AI technologies aim to assist with some of these tasks, including tumor classification, molecular diagnostics, and tumor infiltration.[3–5] Recent advancements have sought to match the diagnostic accuracy of pathologists and to make available to surgeons in the operating room the information that under the current standard of care cannot be obtained until days or weeks postoperatively, such as the presence or absence of key mutations, opening the door to molecularly informed tumor surgeries.[6–8] Key current areas of focus involve augmenting both established standard-of-care procedures, such as cryosection analysis and optimizing fluorescence guidance using 5-aminolevulinic acid (5-ALA) and indocyanine green (ICG), as well as novel technologies, including nanopore sequencing and stimulated Raman histology (SRH).

In this report, we aimed to provide a comprehensive overview of state-of-the-art pathology models that can be used by neurosurgeons intraoperatively and their emerging role in neurosurgical practice (Fig. 1). Generally, these models fall into 3 broad categories: differentiation of lesional and nonlesional tissue for the characterization of surgical margins, diagnostic classification of lesions by tumor type or histological grade, and molecular characterization to determine the WHO classification and underlying biology of a lesion (Table 1).

Surgical Pathology

Cryosection

The gold standard of surgical pathology remains cryosection, an intraoperative method for providing rapid preliminary diagnostics as well as initial treatment strategies and dictating surgical decision-making. Cryosection is performed by freezing and slicing the tissue sample in a cryostat, then staining the slide(s) of interest as needed, generally with an H&E stain. Pathologists then review the slide and report back to the operating surgeon, ideally within 20 minutes (Fig. 2).[9] Because freezing artifacts must be handled robustly, most developments in this field come from models trained specifically on cryosections rather than general pathology foundation models built on H&E-stained, formalin-fixed, paraffin-embedded slides.[10]

One exciting model is LGNet, designed to intraoperatively distinguish primary CNS lymphoma (PCNSL) from non-PCNSL tumors, such as metastatic cancer, glioma, and other lesions, on frozen sections. This distinction is important because aggressive resection can be harmful in patients with PCNSL, which is instead treated primarily with chemotherapy and/or steroids. In one study, LGNet achieved an area under the receiver operating characteristic curve (AUROC) of 0.965 on one external cohort and 0.972 on a separate cohort.[5] Additionally, the model significantly surpassed several of the pathologists used for comparison regarding classification; however, the study also found that pathologists with the assistance of LGNet had the best performance of all.

Another task of interest is the molecular subtyping of glioma, particularly given the emphasis placed on molecular markers in the 2021 WHO CNS tumor criteria. For this task, the context-aware Cryosection Histopathology Assessment and Review Machine (CHARM) was developed and trained on 1524 patients with glioma across 3 different populations for the purpose of distinguishing tumor from benign tissue as well as providing grade information and intraoperative molecular diagnostics.[10] CHARM achieved an AUROC of 0.94 on the held-out test set for differentiation between benign and pathological tissue, 0.84 for distinguishing low-grade (grades 2 or 3) from high-grade (grade 4) tumors, and 0.82 for IDH status. When it came to the IDH-mutant with 1p/19q codeletion (molecular oligodendroglioma), IDH-mutant with intact 1p/19q (molecular astrocytoma), and IDH-wildtype (molecular glioblastoma) classification, CHARM achieved AUROCs of 0.88, 0.90, and 0.93, respectively. CHARM could also classify mutations in ATRX, TP53, CDKN2A/B, CIC, and 1p19q codeletion. Lastly, CHARM's attention maps localized diagnostically relevant regions, indicating a greater likelihood of reliability. CHARM's strong performance on such a wide array of tasks also distinguishes it from many other intraoperative AI models, which often specialize in only one or a few tasks.

Fluorescence

There are multiple fluorescent agents currently in use or under investigation as tools for glioma resection, such as 5-ALA or ICG.[11] After administration of 5-ALA, accumulated protoporphyrin IX emits red fluorescence when excited by 405-nm light, which can be observed directly or through a camera. Notably, however, 5-ALA is less reliable in areas of low infiltration in high-grade gliomas (HGGs) and low-grade gliomas (LGGs). ICG emits near-infrared fluorescence that is less affected by tissue autofluorescence and allows greater tissue penetration but is not visible to the naked eye.[11,12] Despite its very brief half-life of < 180 seconds in healthy patients, ICG accumulates in solid tumors due to the enhanced permeability and retention effect, allowing second window ICG staining to be visible more than 24 hours after infusion.[13] AI models seek to build on the signal created by 5-ALA and ICG by extracting subvisual biological information from samples that are already taken during a standard intraoperative workflow (Fig. 3).

One of the earliest applications of machine learning to tumor fluorescence occurred in 2020, when Leclerc et al. performed a pilot study to identify LGG and the sparsely infiltrated margins of HGG.[14] Using 50 biopsy samples from 10 patients with glioma, they achieved 77% classification accuracy using K-means and Gaussian mixture models. Despite the relatively low accuracy, this paper laid the foundation for future intraoperative fluorescence models.

More recently, Black et al. demonstrated the use of hyperspectral imaging to go beyond simple binary classification to 5-ALA–enabled classification of tumor type,

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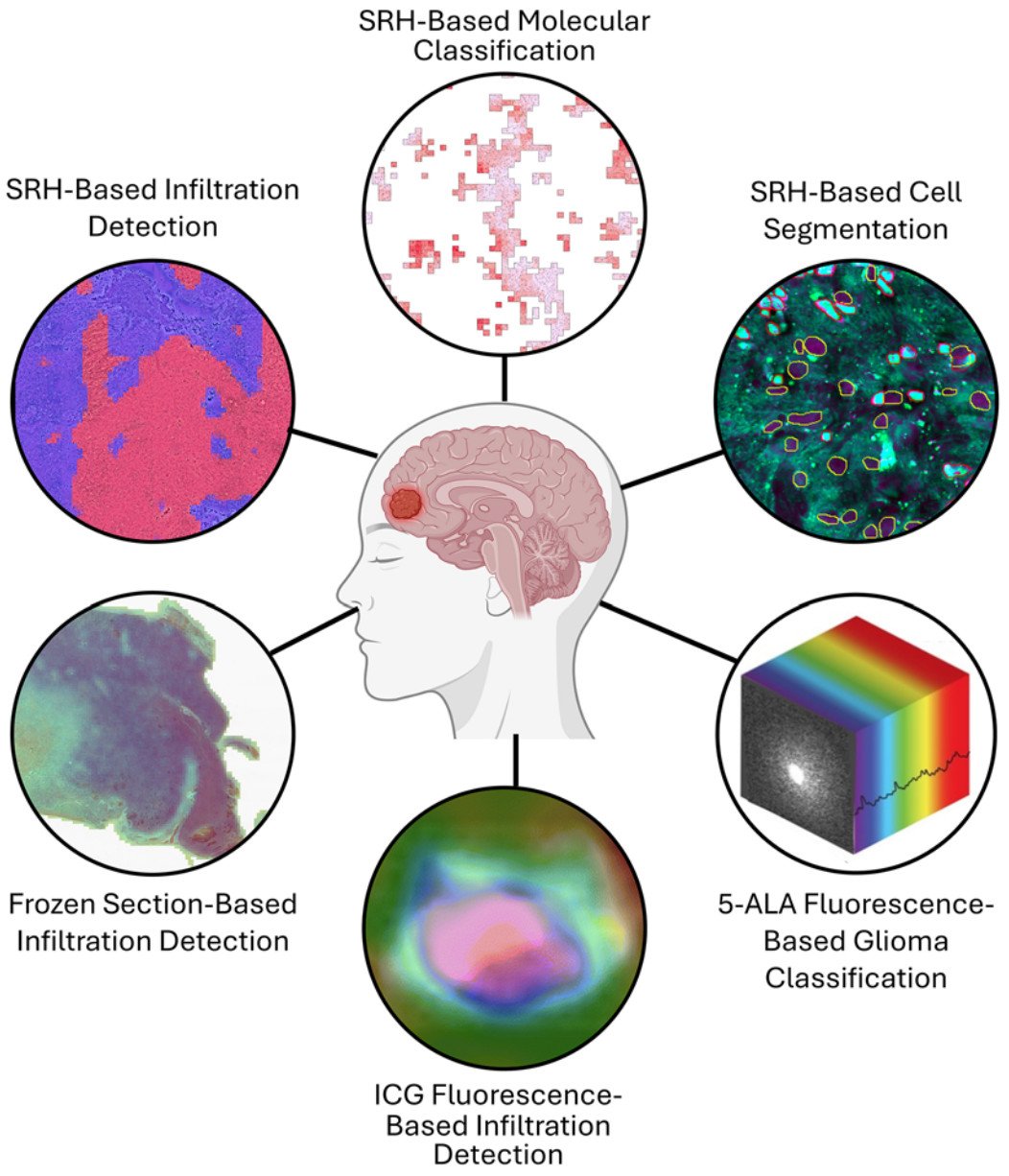


FIG. 1. An overview of the various intraoperative AI models available to assist neurosurgeons during brain tumor resections. 5-ALA image adapted from Black D, Byrne D, Walke A, Liu S, Di Ieva A, Kaneko S, Stummer W, Salcudean T, Molina ES. Towards machine learning-based quantitative hyperspectral image guidance for brain tumor resection. *Commun Med*. 2024;4(1):131. © The authors, published with permission. CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). ICG image adapted from Shen B, Zhang Z, Shi X, Cao C, Zhang Z, Hu Z, Ji N, Tian J. Real-time intraoperative glioma diagnosis using fluorescence imaging and deep convolutional neural networks. *Front Surg*. 2019;6:11. © The authors, published with permission. CC-BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Entire figure created in BioRender. Goff, N. (2026) https://BioRender.com/joud3vr.

histological grade, and IDH mutation by using a set of 891 biopsies from 184 patients.[15] These authors explored some classic machine learning models and some deeper methods like multilayer perceptrons using 5 known fluorophore compounds: 2 protoporphyrin IX states, NADH, lipofuscin, and flavins.[15–17] In another study, Black and colleagues achieved 85.7% accuracy for 3-class tumor margin classification (solid tumor, infiltrating zone, reactive tissue), 96.1% accuracy for WHO grade, 93% accuracy for IDH mutation, and 87.3% accuracy for 10-class tumor type identification.[18] This approach could give surgeons near real-time tissue information during fluorescence-guided resection without major workflow changes.

On the other hand, intraoperative AI-driven applications of ICG are much less developed. To date, only one small pilot study using 1874 samples from 23 patients with glioma has applied machine learning models to ICG fluorescence. A convolutional neural network (CNN) combined with second near-infrared fluorescence imaging (FL-CNN) achieved an AUROC of 0.945 for lesional tissue detection,

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TABLE 1. Characteristics of state-of-the-art pathology models

Modality & Task	Model Name	Authors & Year	Architecture	No. of Patients	Performance
Cryosection					
PCNSL detection	LGNet	Zhang et al., 2024[5]	ResNet50	686 test	AUROC: 0.97
Tumor segmentation	CHARM	Nasrallah et al., 2023[10]	ViT	403 test	AUROC: 0.94
Glioma classification	CHARM	Nasrallah et al., 2023[10]	ViT	403 test	AUROC: 0.90
Fluorescence					
Infiltration detection (5-ALA)	—	Black et al., 2024[18]	MLP	184 train & test	Acc: 85.7%, AUROC: 0.95
Tumor classification (5-ALA)	—	Black et al., 2024[18]	Random forest	184 train & test	Acc: 83.6%, AUROC: 0.98
Tumor detection (SWIG)	FL-CNN	Shen et al., 2021[12]	EfficientNet-B0	23 train & test	Acc: 89.1%, AUROC: 0.945
Nanopore sequencing					
Methylation-based classification	ROBIN	Deacon et al., 2025[21]	NanoDx, crossNN, Sturgeon	50 test	Acc: 90%
SRH					
Infiltration detection	FastGlioma	Kondepudi et al., 2025[34]	ResNet34	220 test	AUROC: 0.921
Glioma classification	DeepGlioma	Hollon et al., 2023[33]	ResNet50	153 test	Acc: 91.5%, AUROC: 0.926
PCNSL detection	RapidLymphoma	Reinecke et al., 2025[41]	ResNet50	168 test	Acc: 97.8%, AUROC: 0.987
Cell nuclear detection	ELUCIDATE	Bhattacharya et al., 2025[42]	Mask R-CNN	35 CV	AP50: ≈0.85

Acc = accuracy; AP50 = average precision with an intersection over union cutoff of 0.5; CV = cross-validation; Mask R-CNN = Mask region-based CNN; MLP = multilayer perceptron; SWIG = second window indocyanine green; ViT = Vision Transformer.

outperforming both neurosurgeons and white light–based AI models. FL-CNN also achieved AUROCs of 0.810 in differentiating between LGG and HGG and 0.688 for Ki-67 classification.[12] These results, particularly those for the lesional versus nonlesional tissue task, are comparable to those of Black and colleagues. ICG-based intraoperative AI models have not been well studied since.

Nanopore Sequencing

To obtain DNA methylation results, traditional methods, such as Illumina EPIC microarrays, tend to be expensive and have long turnaround times. To mitigate this, nanopore sequencing was designed to be an alternative method; however, even this method only yields sparse profiles within intraoperative time constraints (about 100–400 mb in 60–90 minutes; Fig. 4). One of the first well-validated AI models for DNA methylation classification based on nanopore sequencing was Sturgeon, which correctly identified 45 of 50 patients in a retrospective sample on the basis of data collected over 20- to 40-minute intervals.[19] Prospective validation across 25 surgeries resulted in 18 classified correctly, while the remaining 7 were classified as not having enough information rather than as incorrect.

To handle multiple DNA methylation profiling platforms, crossNN was developed and incorporated explainability via interpretable feature weights.[20] Predictions could be performed in seconds and thus were reliably obtained in the operating room. Using cross-validation, 96.11% ± 0.86% overall accuracy was achieved across all folds at the methylation class (MC) level and 99.07% ± 0.21% accuracy at the methylation class family (MCF) level. When independently validated, overall accuracy on the MC and MCF level was 0.83 and 0.88, respectively.

Since Sturgeon and crossNN, more models have been developed, focusing on ensembling AI methods together to improve accuracy. For example, Rapid Nanopore Brain Intraoperative Classification (ROBIN) uses Sturgeon, crossNN, and a random forest to improve diagnostic confidence when mutual agreement occurs.[21] Across 50 intraoperative cases, diagnosis was achieved prospectively in under 2 hours and reached concordance with the final diagnosis in 90% of cases. It also reached high confidence in 76% of prospective cases and thus intraoperative reliability. MGMT methylation predictions were also concordant with the final diagnosis in more than 95% of prospective cases, and IDH mutation was correctly identified in 9 of 10 oligodendroglioma and astrocytoma cases.

Stimulated Raman Histology

SRH is a rapid, label-free, nondestructive optical imaging technique. Based on Raman spectroscopy, which uses inelastic light scattering to collect chemical spectra from samples, SRH collects channels in 1000 pixel-wide strips at Raman shifts of 2845 cm^{-1} (CH_2 bonds; lipids) and 2930 cm^{-1} (CH_3 bonds; proteins and nucleic acids), with a third channel created using pixel-wise subtraction of the other two channels.[22,23] These strips are then processed and concatenated into a single whole-slide image, which is then recolored for display as a virtual H&E image.[24] SRH has been shown to have high concordance with frozen section and permanent pathology slides while having a far faster turnaround time and allowing surgeons to personally review images from the operating table (Fig. 5).[24–28] Our group has been involved in the development of many of the SRH models presented here, like DeepGlioma and RapidLymphoma.

Brain Tumor Segmentation

One of the most compelling use-cases for SRH is in

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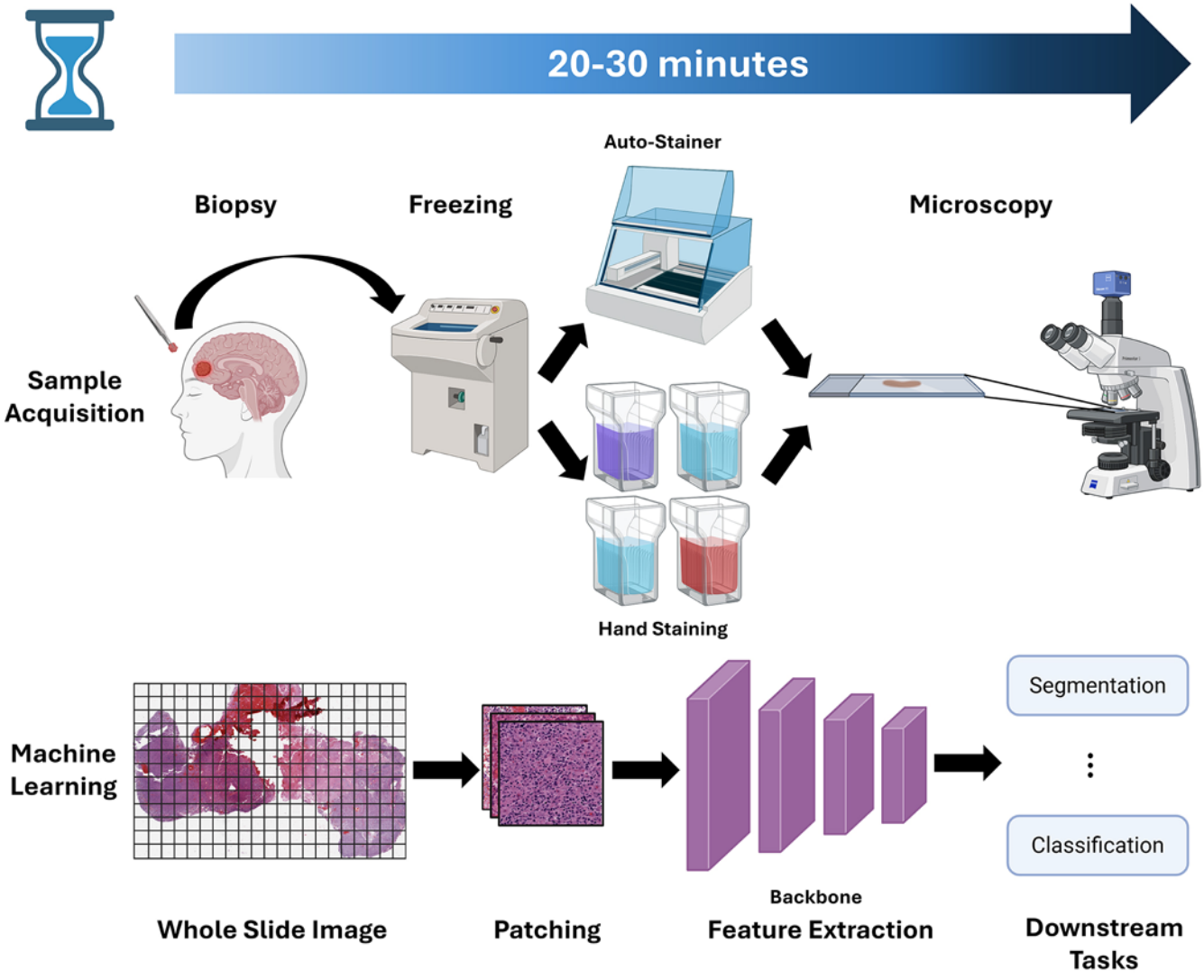


FIG. 2. An overview of the acquisition workflow and AI models available for cryosection. Following biopsy, the sample is frozen via cryostat, then either manually or automatically stained using an H&E stain. Microscopic images are then collected, patched, and passed through a feature extractor. The resulting features are used for downstream tasks, such as tumor segmentation or classification. H&E image courtesy of The Cancer Genome Atlas, National Cancer Institute and National Human Genome Research Institute, National Institutes of Health, US Department of Health and Human Services. Entire figure created in BioRender. Goff, N. (2026) https://BioRender.com/jrecq4m.

determining whether tissue is lesional or nonlesional. The earliest SRH model, described in 2017, could differentiate between lesional and nonlesional tissue at the patient level using leave-one-out cross-validation with an AUROC of 0.984.[24] This was followed by a similar classifier for pediatric brain tumors, achieving a cross-validation AUROC of 0.970.[29] Subsequent models effectively segmented areas of lesional and nonlesional tissue in both newly diagnosed and recurrent gliomas, differentiating normal tissue and pseudoprogression from glial and nonglial tumors.[30–32] These segmentation models became the basis for future molecular and subtype classification models, such as DeepGlioma.[33]

FastGlioma, a foundation model for SRH imaging and the current state of the art, took the next logical step forward: instead of simple binary classification of lesional and nonlesional tissue, it was trained to identify the degree of glioma infiltration of tissue on a slide image on a scale of 0–3. Tested on a prospective cohort of 220 patients, FastGlioma achieved an AUROC of 0.921 in distinguishing the 4 classes. Furthermore, in differentiating between normal (infiltration score = 0) and densely infiltrated (infiltration score = 3) tissue, FastGlioma achieved an AUROC of 0.981, outperforming current standard-of-care methods, that is, FLAIR positivity (AUROC = 0.763), contrast enhancement (0.718), and 5-ALA fluorescence (0.890).[34]

These results demonstrate SRH's ability to act as an adjunct during resections at the tumor margin. Extent of resection and residual tumor burden are both strong

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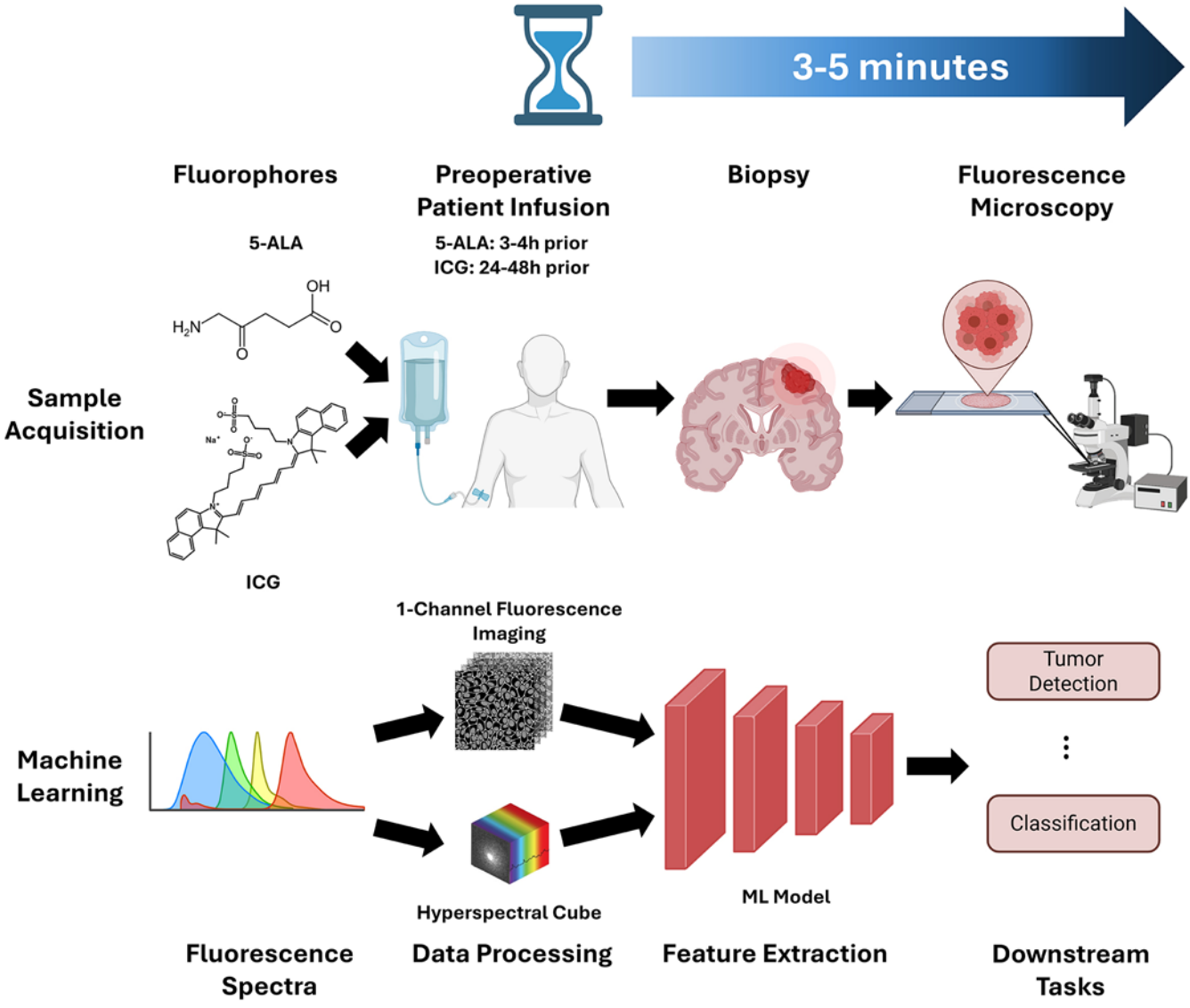


FIG. 3. An overview of the acquisition workflow and machine learning (ML) models available for intraoperative fluorescent agents, such as 5-ALA and ICG. The fluorescent agent is first administered to the patient via intravenous infusion 3–4 (5-ALA) or 24–48 (ICG) hours before surgery. Following biopsy, the sample is examined by fluorescence microscopy. The resulting fluorescence spectra are used to create single-channel images based on intensity or hyperspectral cubes, which are then passed to an ML model for downstream tasks, such as tumor detection and classification. h = hours. Hyperspectral cube image adapted from Black D, Byrne D, Walke A, Liu S, Di Ieva A, Kaneko S, Stummer W, Salcudean T, Molina ES. Toward machine learning-based quantitative hyperspectral image guidance for brain tumor resection. *Commun Med*. 2024;4(1):131. © The authors, published with permission. CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Entire figure created in BioRender. Goff, N. (2026) https://BioRender.com/x18zqwu.

prognostic markers in diffuse glioma, and safe maximal resection is the current standard of care.[6–8,23] A core advantage of SRH lies in giving surgeons rapid margin feedback without breaking sterility or relying on time-sensitive markers like intraoperative MRI or 5-ALA, which it also outperforms.[34,35]

Brain Tumor Classification

There have been multiple models that leverage SRH to classify different types of brain tumors. Early models focused on relatively simple tasks, such as differentiation among normal tissue, low-grade glial tumor, high-grade glial tumor, and nonglial tumor.[24] As more specimens were collected, SRH models were used in more powerful applications, differentiating 13 different histological categories covering both nonlesional (white matter, gray matter, gliosis, etc.) and lesional (glioma, medulloblastoma, meningioma, etc.) entities with high accuracy (94.6%), and were noninferior to conventional H&E-based histological analysis.[30]

Since the publication of the 2021 CNS tumor classification guidelines, molecular markers, rather than histological appearance, have become the basis for glioma classifica-

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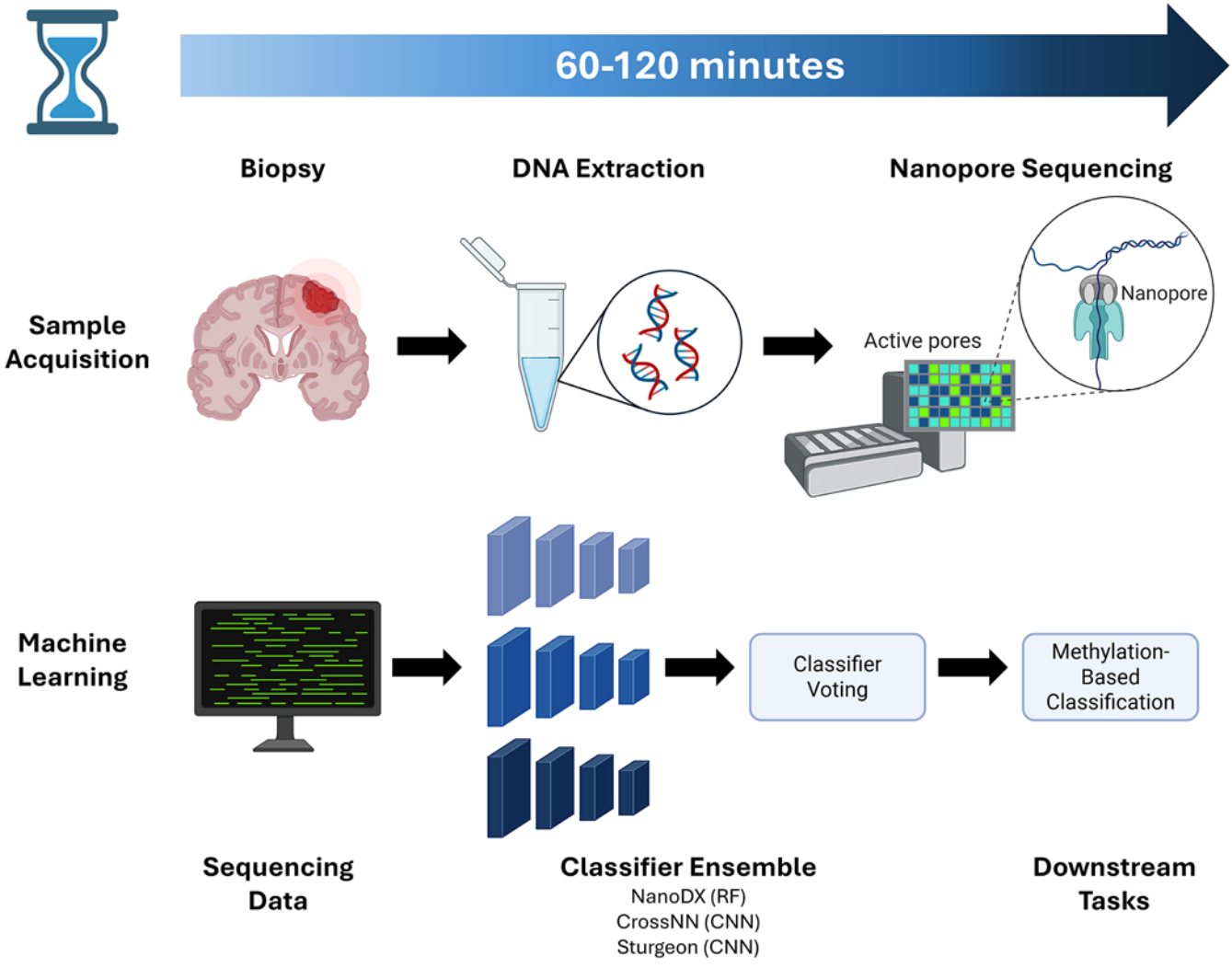


FIG. 4. An overview of the nanopore sequencing and methylation classification process. Following biopsy, the sample undergoes DNA extraction and is then sequenced using the PromethION sequencer (Oxford Nanopore Technologies). Within an hour, ROBIN passes available sequencing data to a 3-classifier ensemble consisting of a random forest (RF) classifier (NanoDX) and 2 CNNs (crossNN, Sturgeon). The classifiers then vote to determine the final MC output. Created in BioRender. Goff, N. (2026) https://BioRender.com/d3fdm2y.

tion.[36] More recent models for intraoperative classification of brain tumors have therefore shifted focus from purely glioma subtype classification to molecular classification, which then informs the final AI-based diagnosis. The first of these models, DeepGlioma, used a transformer-based architecture that predicted the molecular status and subtype of gliomas in under 90 seconds, with a subtype classification accuracy of 91.5% on a multi-institutional test set.[33]

SRH has also been shown to perform well in other difficult neuro-oncological classification tasks, such as the differentiation of PCNSL from other CNS entities, including glioma and meningioma.[37–40] Utilizing a self-supervised learning method on patched SRH image features and a prospective test cohort of 160 patients, RapidLymphoma achieved a balanced accuracy of 97.81% in differentiating PCNSL from a wide variety of other diagnoses, such as IDH-wildtype and IDH-mutant gliomas, metastases, neuroepithelial tumors, and radiation-induced necrosis.[41]

Cell Identification

The study of individual cells has been a relatively new avenue in SRH. Recent models have improved the quality of cell segmentations on SRH images, enabling the development of cell-level models.[42] For example, a recent study showed that there were differences in nuclear area, perimeter, and eccentricity among IDH-wildtype, IDH-mutant, and nontumor cells.[43] Another study showed that a cell-based model achieved 90.28% overall accuracy for glioma subtype classification, demonstrating performance similar to patch-based models such as DeepGlioma.[44]

Future of Intraoperative AI for Glioma Surgery

AI stands to play a major role in the future of neuro-oncology, expanding the breadth of information available to surgeons during brain tumor resection. Models that pro-

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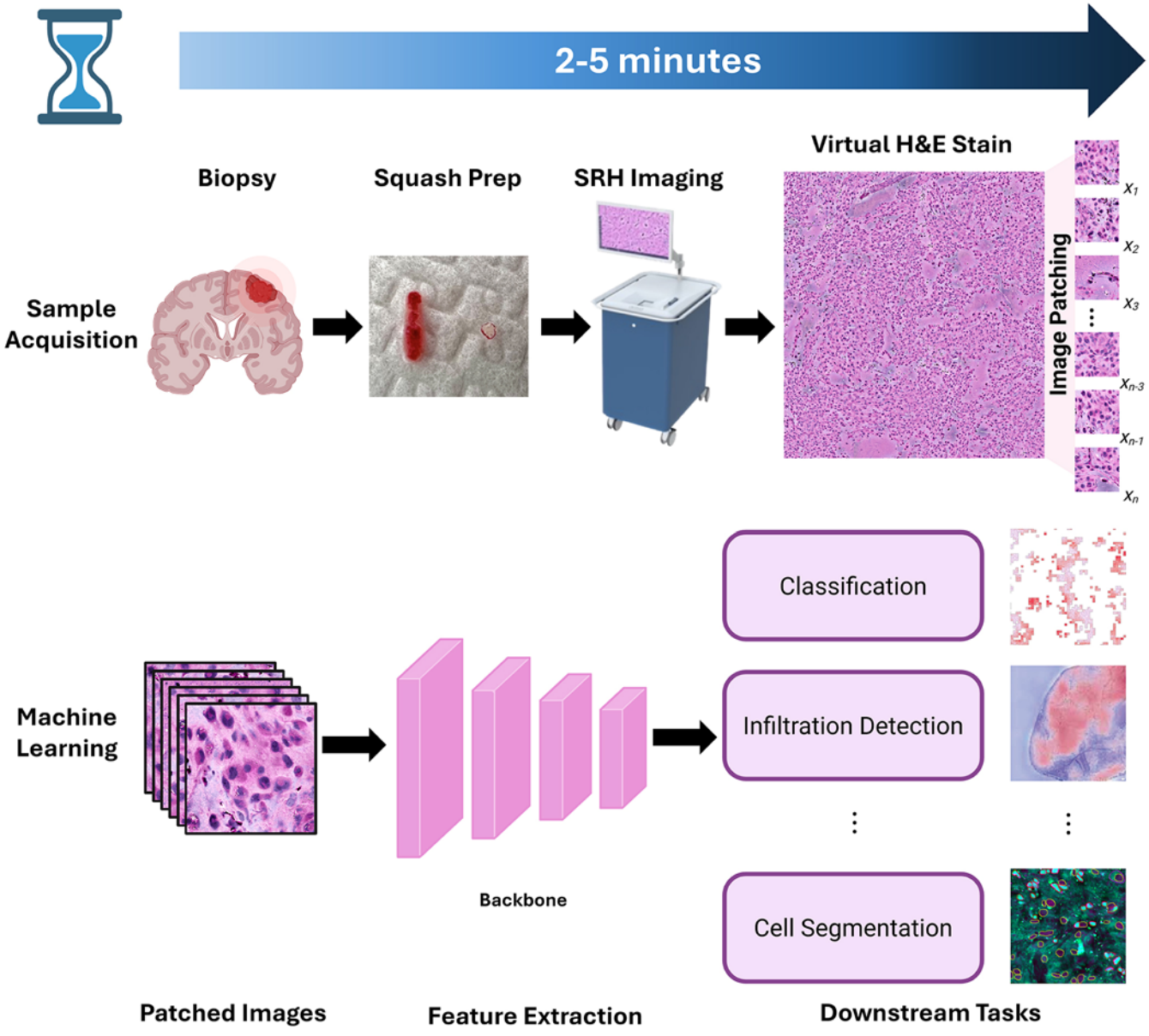


FIG. 5. An overview of the acquisition workflow and AI models available for SRH imaging. Following biopsy, the sample is placed on a microscopy slide for a squash preparation. The slide is then inserted into the SRH imaging system, which produces an image with a virtual H&E stain. This image is then patched and passed through a feature extractor. Image features are used for downstream tasks, such as tumor classification, infiltration detection, and cell segmentation. Created in BioRender. Goff, N. (2026) https://BioRender.com/jucwc07.

vide real-time insight into a tumor's molecular and histological characteristics could allow surgical decision-making to be guided by tumor biology in ways not previously possible. Although the US Food & Drug Administration (FDA) has cleared several pathology-related AI-enabled medical devices, including both prognostic risk stratification tools and algorithms that detect or diagnose lesional tissue, none are indicated for neuro-oncology applications or, more importantly, designed for intraoperative use.[45] This gap highlights an important opportunity for intraoperative AI to advance more personalized surgical care for patients with brain cancer. However, many hurdles remain to be cleared before the widespread use of intraoperative models is safe and feasible.

The most crucial barrier to the adoption of the above-described intraoperative technologies is a lack of rigorous and reproducible data for regulatory approval beyond experimental usage. Current evidence on the intraoperative utility of certain models, such as those that provide information about the degree of infiltration or molecular status of the tissue, is encouraging but insufficient for clinical translation. In recent years, more models have been examined with simulated clinical trials, usually single-arm noninferiority trials that show that the model is noninferior

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to the current standard of care for their respective tasks.[34] However, these studies are often underpowered, relying on narrowly controlled experimental settings (e.g., curated datasets for model training, small external datasets for model validation). Furthermore, many studies have been performed without external validation, such as with an extra-institutional or a prospective test set, leading to decreased reliability of reported metrics. Researchers should seek to validate the clinical utility of extant models with rigorous randomized trials focused on predefined clinical performance endpoints such as extent of resection, diagnostic accuracy, and differences in survival and neurological morbidity. Although these types of studies are difficult to design and perform, without randomized clinical trials demonstrating noninferiority to or improvement over the current standard of care, widespread adoption will be limited.

Regulatory approval and further validation of these extant models via randomized trials are particularly crucial, as the information they provide could be invaluable to neurosurgeons intraoperatively and has the potential to alter the surgical plan. Though margin detection for extent-of-resection guidance is certainly useful, it is far from the only model that could provide intraoperative utility. For example, among gliomas, multiple studies have demonstrated that extent of resection is a stronger prognostic marker in IDH-mutant, 1p19q-intact astrocytomas than in IDH-mutant, 1p19q-codeleted oligodendrogliomas or IDH-wildtype glioblastomas.[6–8,46] In eloquent areas of the brain, this information could augment surgical decision-making, helping to optimize surgical outcomes and decrease neurological morbidity. Similarly, in PCNSL, the standard of care is stereotactic needle biopsy and a combination of chemotherapy, radiation, and autologous stem cell transplant rather than resection because of a poor clinical response to surgery and the high risk of neurological morbidity.[37,47,48] Although no published trials have documented the clinical utility of any classification models, RapidLymphoma has demonstrated noninferiority to frozen section pathology on 25 patients with PCNSL in a prospective test set.[10,33,37,41]

Additionally, systematic comparisons between technologies, including evaluation of the human-AI interaction, are needed before widespread adoption to assess performance under realistic intraoperative conditions. For example, Zhang et al. found that LGNet, which differentiates PCNSL from non-PCNSL entities using cryosection, performed best when used by neuropathologists as an adjunct rather than as a replacement for human expertise.[5] Though future work should continue to compare model performance and human performance separately, as clinical deployment becomes more commonplace, researchers should consider investigating augmented human performance as well.

Ultimately, regulatory frameworks are evolving in response to the rapid growth of AI in medicine and the distinct challenges posed by such tools. Recent FDA initiatives have emphasized a total product lifecycle approach that extends beyond initial clearance to include dataset quality, bias assessment, transparency, cybersecurity, human factors testing, postmarket surveillance, and oversight of model updates over time.[49] In particular, the FDA's emerging framework for predetermined change control plans reflects the recognition that future AI systems may continue to evolve after deployment, underscoring the need for regulatory mechanisms that preserve reproducibility, reliability, and patient safety over time. As AI-enabled devices become more common, demonstrating evidence of clinical utility and generalizability across institutions and diverse patient populations will be essential.

Beyond regulatory challenges, there are further barriers to the adoption of this technology into clinical practice. First and foremost, the use of AI represents a shift in current clinical workflows, where there is likely inertia that will affect its incorporation. Institutional workflows and policies are difficult to change, particularly when the status quo has been proven to be effective. There is also a cost associated with the hardware and software necessary for the models described above; although cryosection is relatively standard, nanopore sequencing involves technical challenges and SRH equipment costs that may be prohibitive for adoption outside of large hospitals. However, as AI becomes increasingly adopted by society at large and computing costs continue to decrease, the hope is that the impact of these barriers will be decreased.

There are also many outstanding questions that may be addressed by intraoperative AI models. For example, few if any of these models are multimodal. The fusion of pre- or intraoperative imaging with pathology data allows more advanced models to place information about the tumor microenvironment in the context of the patient's and the tumor's anatomy. In clinical practice, we generate a large suite of data types—patient history, physical examination, imaging, pathology, etc.—and AI can be used to detect patterns in these multimodal datasets. Especially for questions that have proved difficult to model using a single modality, such as survival in glioma, multimodal models may provide a path forward.

Conclusions

Since the publication of the 2021 WHO CNS tumor guidelines, an understanding of the molecular profile is critical for accurate diagnostics. The recent development of rapid AI-based intraoperative pathology technologies is transforming the current landscape to provide information in real time, representing a broader shift to more data-driven approaches in neurosurgery. While further prospective validation is needed for these technologies to become the standard of care, a reduced dependence on intraoperative neuropathology allows diagnostics to be rapidly obtained, regardless of institutional resources. Each technology is designed to meet a specific intraoperative need, from rapid methylation profiling to real-time margin assessment to molecular characterizations. The continued adoption of these technologies into surgical workflows has the potential to bring neuro-oncology into the age of real-time biologically informed interventions.

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Author Contributions

Conception and design: Orringer. Acquisition of data: Goff, Park, Springer, Orringer. Analysis and interpretation of data: Goff, Park, Orringer. Drafting the article: Goff, Markert, Park, Chen. Critically revising the article: Goff, Markert, Chen, Orringer. Reviewed submitted version of manuscript: Markert, Park, Springer, Chen, Orringer. Approved the final version of the manuscript on behalf of all authors: Goff. Administrative/technical/material support: Chen, Orringer.

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